Exact flat-noise transition rays from Hadamard involutions in four-level systems

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For a four-level system prepared and measured in one fixed basis, a unitary U determines all sixteen transition probabilities $|U_{ij}|^2$. We construct a noncirculant family for which every convex mixture of this probability matrix with the uniform matrix is itself generated by a single unitary. If F is a real normalized Hadamard matrix and P is a real orthogonal projector, then $U = F(I - 2P)$ admits the exact witness $V_\lambda = \sqrt{1-\lambda}\,U + i\sqrt{\lambda}\,F$ for every $0 \le \lambda \le 1$. We prove that the projector condition is also necessary within this particular real quadrature ansatz. An explicit rational rank-two projector gives a nonzero exact Jacobian minor, showing that the resulting noncirculant transition matrices have a locally four-dimensional image. The construction also supplies a gate with a fixed Hermitian generator and a tunable evolution angle. It reproduces the population statistics of a depolarizing mixture, without equating the corresponding quantum channels on coherent inputs. The unrestricted four-dimensional unistochastic star-shapedness question remains open.

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I. INTRODUCTION AND PHYSICAL SETTING

Let $\{|j\rangle\}_{j=1}^{4}$ be a fixed orthonormal preparation and measurement basis. An applied $U \in U(4)$ produces the conditional probabilities

$$B(U)_{ij} = |U_{ij}|^2 = \Pr(i \text{ measured} \mid j \text{ prepared}), \qquad W_{ij} = \tfrac{1}{4}. \tag{1}$$

The probability matrix $B(U)$ is unistochastic. Complete depolarization after U, mixed with weight $\lambda \in [0, 1]$, has the fixed-basis transition matrix

$$M_\lambda(U) = (1 - \lambda)B(U) + \lambda W. \tag{2}$$

The general question is whether $M_\lambda(U)$ is unistochastic for *every* $U \in U(4)$ and every λ on the closed interval. It is distinct from asking whether a unitary channel equals the noisy channel on arbitrary input states.

Permutation-to-flat rays have explicit unitaries, and certain two-permutation triangles were also constructed by Rajchel, Gąsiorowski, and Życzkowski.[1] The flat matrix is on the boundary of the order-four unistochastic set,[2] so a local argument based on an interior ball is unavailable. Rajchel-Mieldzioć *et al.* proved star-shapedness for the larger bracelet and factorisable sets and, separately, the unistochastic property for the order-four circulant subclass.[3] A later discussion still treats the unrestricted problem as conjectural.[4] We give a sufficient family that contains noncirculant rays and has a locally four-dimensional probability image. It does not resolve the unrestricted question.

All probabilities and λ are dimensionless. There is no perturbative expansion, spatial boundary condition, or imposed control time. Throughout, columns label prepared basis states and rows label measured outcomes.

II. AN EXACT QUADRATURE CONSTRUCTION

A real normalized Hadamard matrix of order n is an $F \in O(n)$ with $F_{ij}^2 = 1/n$. An orthogonal projector is a real matrix $P = P^T = P^2$. Its reflection

$$S = I - 2P \tag{3}$$

is symmetric and orthogonal. The following statement is given at general order to expose the mechanism; the new explicit family and dimensional certificate below concern $n = 4$.

Proposition. *If a real normalized Hadamard F of order n exists and P is any real orthogonal projector, set $U = F(I - 2P)$. Then for every $\lambda \in [0, 1]$,*

$$V_\lambda = \sqrt{1-\lambda}\, U + i\sqrt{\lambda}\, F \tag{4}$$

is unitary and satisfies

$$|(V_\lambda)_{ij}|^2 = (1-\lambda)|U_{ij}|^2 + \frac{\lambda}{n} \quad \textit{for all } i, j. \tag{5}$$

Proof. Put $a = \sqrt{1-\lambda}$ and $b = \sqrt{\lambda}$. Since $S^T = S$ and $S^2 = I$, both F and $U = FS$ are real orthogonal and

$$UF^T = FSF^T = FU^T. \tag{6}$$

Consequently,

$$V_\lambda V_\lambda^\dagger = (a^2 + b^2)I + iab(FU^T - UF^T) = I. \tag{7}$$

Each U_{ij} and F_{ij} is real, so the real and imaginary parts in Eq. (4) add in quadrature. This gives Eq. (5) entry by entry. The construction includes $V_0 = U$ and $V_1 = iF$ directly. □

The same condition exactly characterizes this *particular ansatz*. Let $F, U \in O(n)$ with F flat and fix any interior $0 < \lambda < 1$. Equation (7) shows that $aU + ibF$ is unitary if and only if $FU^T = UF^T$. On multiplying by F^T and F, this is equivalent to $S_0 = F^T U$ being symmetric. Since S_0 is already orthogonal, $S_0^2 = I$ and $P = (I - S_0)/2$ is an orthogonal projector. Thus $U = F(I - 2P)$. This converse is confined to the fixed real quadrature ansatz; other phase choices might realize a ray even when the ansatz fails.

III. A NONCIRCULANT FOUR-LEVEL FAMILY

Take the normalized real Hadamard matrix

$$F = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{8}$$

For a rank-one illustration, set $v = (1, 2, 3, 5)^T$ and $P_1 = vv^T/39$. Equation (3) gives the exact orthogonal matrix

$$U_1 = F(I - 2P_1) = \frac{1}{78} \begin{pmatrix} 17 & -5 & -27 & -71 \\ 45 & -27 & 57 & -9 \\ 49 & 59 & -9 & 11 \\ 37 & -43 & -45 & 29 \end{pmatrix}. \tag{9}$$

Its first probability row is $(289, 25, 729, 5041)/6084$, with four distinct entries. Its second row is $(2025, 729, 3249, 81)/6084$. The two rows have different entry multisets. Every row of a circulant matrix has the same entry multiset, a property preserved by row and column relabeling. Therefore $B(U_1)$ is not equivalent to a circulant matrix under such relabeling. Four distinct entries in one row also separate it from permutation-to-flat rays and the cited two-permutation triangles, which have at most two and three values per row, respectively. For example, the entire ray at $\lambda = 1/2$ has the exact witness $(U_1 + iF)/\sqrt{2}$; Eq. (5) applies to all sixteen entries and all λ.

The rank-two component is more than a single example. Let $w = (2, -1, 0, 0)^T$, so $v^T w = 0$, and set $P_2 = P_1 + ww^T/5$. In rational form,

$$P_2 = \frac{1}{195} \begin{pmatrix} 161 & -68 & 15 & 25 \\ -68 & 59 & 30 & 50 \\ 15 & 30 & 45 & 75 \\ 25 & 50 & 75 & 125 \end{pmatrix}. \tag{10}$$

Direct multiplication gives $P_2^2 = P_2$ and $\operatorname{tr} P_2 = 2$. Define $\mathcal{B}(P) = B(F(I - 2P))$. The rank-two projectors form the real Grassmannian $\mathrm{Gr}(2, 4)$, of dimension four. To test whether $\mathcal{B}$ retains that local dimension, let $K_{ab} = E_{ab} - E_{ba}$ for $a < b$ and differentiate the orthogonal orbit $P(t) = e^{tK_{ab}} P_2 e^{-tK_{ab}}$. With $U_2 = F(I - 2P_2)$,

$$\dot{P} = [K_{ab}, P_2], \qquad \dot{U} = -2F\dot{P}, \qquad \dot{\mathcal{B}}_{ij} = 2(U_2)_{ij}\dot{U}_{ij}. \tag{11}$$

Using output coordinates $(B_{11}, B_{12}, B_{13}, B_{21})$ and orbit directions $(K_{12}, K_{13}, K_{14}, K_{23})$, the exact determinant is

$$\det \left[\frac{\partial(B_{11}, B_{12}, B_{13}, B_{21})}{\partial(K_{12}, K_{13}, K_{14}, K_{23})} \right]_{P_2} = \frac{211601016}{7843564625} \neq 0. \tag{12}$$

Here the derivative notation denotes the four orbit parameters in the stated order. The inverse-function theorem gives a locally four-dimensional probability image. The rows of $\mathcal{B}(P_2)$ also have unequal entry multisets: its first row contains 5041/152100, whereas its second row is $(6561/16900, 1089/16900, 361/676, 9/676)$ and contains no such value. This inequality persists in a neighborhood. Thus the construction supplies a locally four-dimensional set of noncirculant transition matrices whose complete flat-noise rays have explicit unitary witnesses. The rational certificate in Eq. (12) is reproducible from Eq. (11) and the accompanying exact-arithmetic script.

IV. POPULATION-LEVEL GATE INTERPRETATION AND LIMITS

With $\theta = \arcsin\sqrt{1-\lambda} \in [0, \pi/2]$, Eq. (4) can be rewritten as

$$V_\lambda = iF\exp(-i\theta S), \qquad S = I - 2P. \tag{13}$$

Indeed, $S^2 = I$ gives $i\exp(-i\theta S) = \sin\theta\, S + i\cos\theta\, I$, with $\sin\theta = \sqrt{1-\lambda}$ and $\cos\theta = \sqrt{\lambda}$. A fixed dimensionless Hermitian involution S therefore generates the variable part of the path; a physical Hamiltonian is $H = \hbar S/t_0$ for any chosen nonzero reference time t_0, with evolution time $t = t_0\theta$. The flat gate F is applied afterward. No claim is made about a prescribed hardware time or control constraints.

For comparison, the noisy channel is

$$\Phi_{\lambda,U}(\rho) = (1-\lambda)U\rho U^\dagger + \lambda\,\tfrac{I_4}{4}\operatorname{Tr}\rho. \tag{14}$$

Both this map and unitary conjugation by V_λ are completely positive and trace preserving. Their sixteen fixed-basis preparation/measurement probabilities agree by Eq. (5); the channels generally differ for coherent inputs and when tested with a reference system. At $\lambda = 1$, for example, Eq. (14) erases every input while iF preserves purity. The result concerns population statistics, not channel replacement in general quantum protocols.

V. METHODS AND REPRODUCIBILITY

The proof of Eqs. (7) and (5) is algebraic and exact. The rational matrices and Jacobian minor were computed with SymPy 1.14.0 under Python 3.12.14. The accompanying

`hadamard_involution.py` constructs F, P_1, and P_2 from the displayed rational vectors, checks orthogonality and all-parameter symbolic identities, and computes Eq. (12) using six elementary skew generators. The script and instructions permit regeneration without numerical tolerances.

OpenAI Codex (OpenAI; model `gpt-5.6-sol`, accessed September 2026) assisted in proposing the quadrature construction, deriving and checking the symbolic steps, analyzing primary literature, and drafting this manuscript. It was used to accelerate exact symbolic exploration and source comparison. The displayed identities and rational certificate permit direct checks without relying on the model's output. Critical scientific review and accountability by the listed human author remain necessary before submission.

VI. CONCLUSION

Real Hadamard reflections give exact unitary witnesses for the full closed flat-noise ray of a four-level transition matrix. The within-ansatz converse identifies the relevant real orthogonal involutions, and a rational rank-two example certifies a locally four-dimensional noncirculant image. The gate interpretation applies to fixed-basis populations only. These statements give restricted constructive progress while leaving the question for arbitrary $U \in U(4)$ unresolved.

AUTHOR DECLARATIONS

Conflict of interest. A conflict-of-interest declaration requires confirmation by the human author before submission.

DATA AVAILABILITY

All exact matrices and the script used to regenerate the symbolic certificate are included with the local source package. No experimental data were used.

REFERENCES

[1]G. Rajchel, A. Gąsiorowski, and K. Życzkowski, "Robust Hadamard matrices, unistochastic rays in Birkhoff polytope and equi-entangled bases in composite spaces," Mathematics in Computer Science **12**, 473–490 (2018).

[2]I. Bengtsson, Å. Ericsson, M. Kuś, W. Tadej, and K. Życzkowski, "Birkhoff's polytope and unistochastic matrices, $N = 3$ and $N = 4$," Communications in Mathematical Physics **259**, 307–324 (2005).

[3]G. Rajchel-Mieldzioć, K. Korzekwa, Z. Puchała, and K. Życzkowski, "Algebraic and geometric structures inside the Birkhoff polytope," Journal of Mathematical Physics **63**, 012202 (2022).

[4]R. Salazar, J. Czartowski, R. Ravell Rodríguez, G. Rajchel-Mieldzioć, P. Horodecki, and K. Życzkowski, "Quantum resource theories beyond convexity," (2026), arXiv:2405.05785v6 [quant-ph].